# The Water Cycle, Atmosphere, and Cryosphere

OCN 623 – Chemical Oceanography

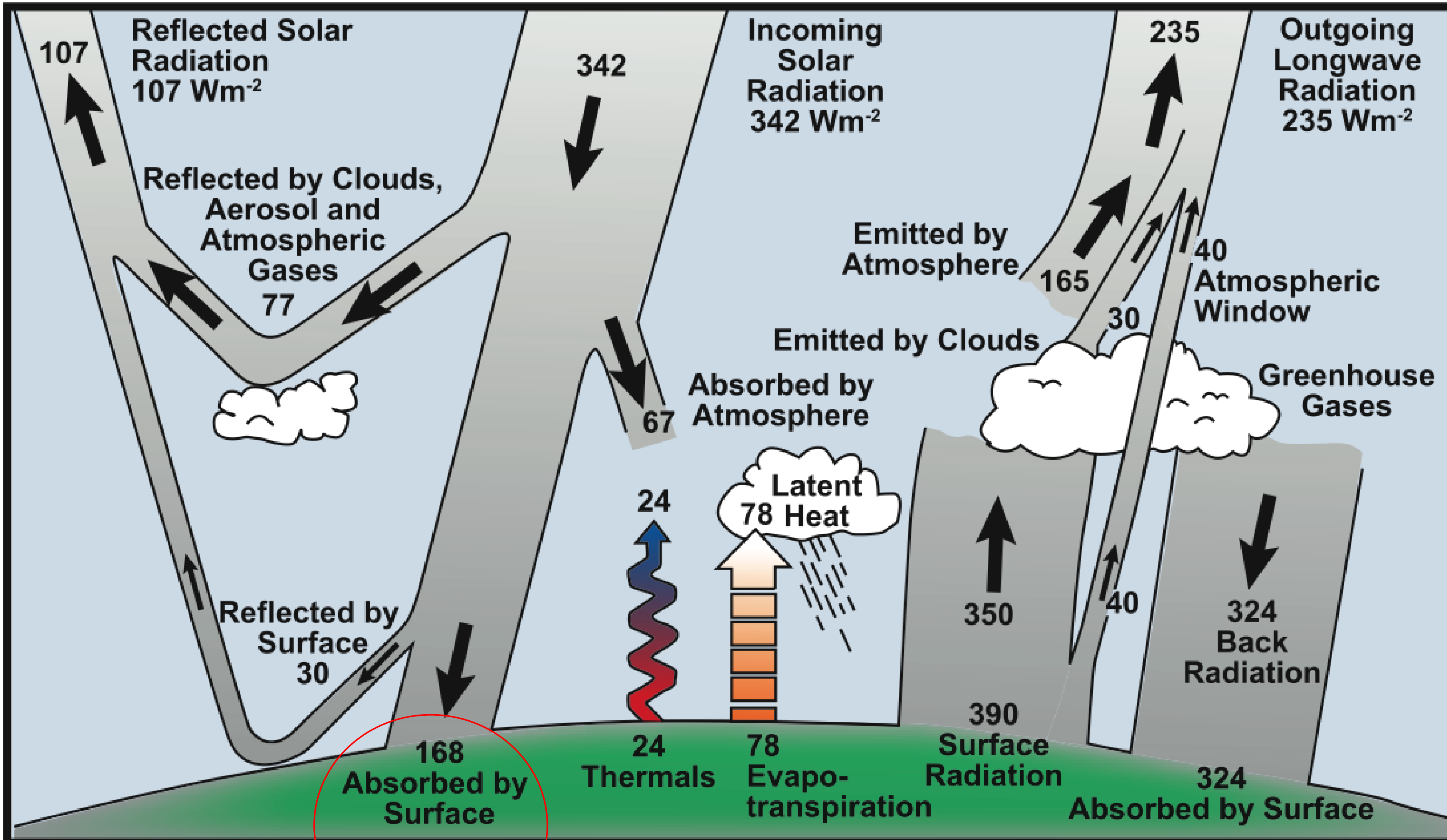
# Student Learning Outcomes (SLOs)

**At the completion of today's section, students should be able to:**

1. **Summarize components of the Earth's solar Energy budget**
2. **Illustrate a generalized atmospheric circulation model & relate to surface water salinity**
3. **Describe the role of cryosphere in global energy budget**
4. **Understand how ice changes in Antarctica have different implications for the globe than the Arctic**

The average Albedo (reflectivity) of Earth is 0.3

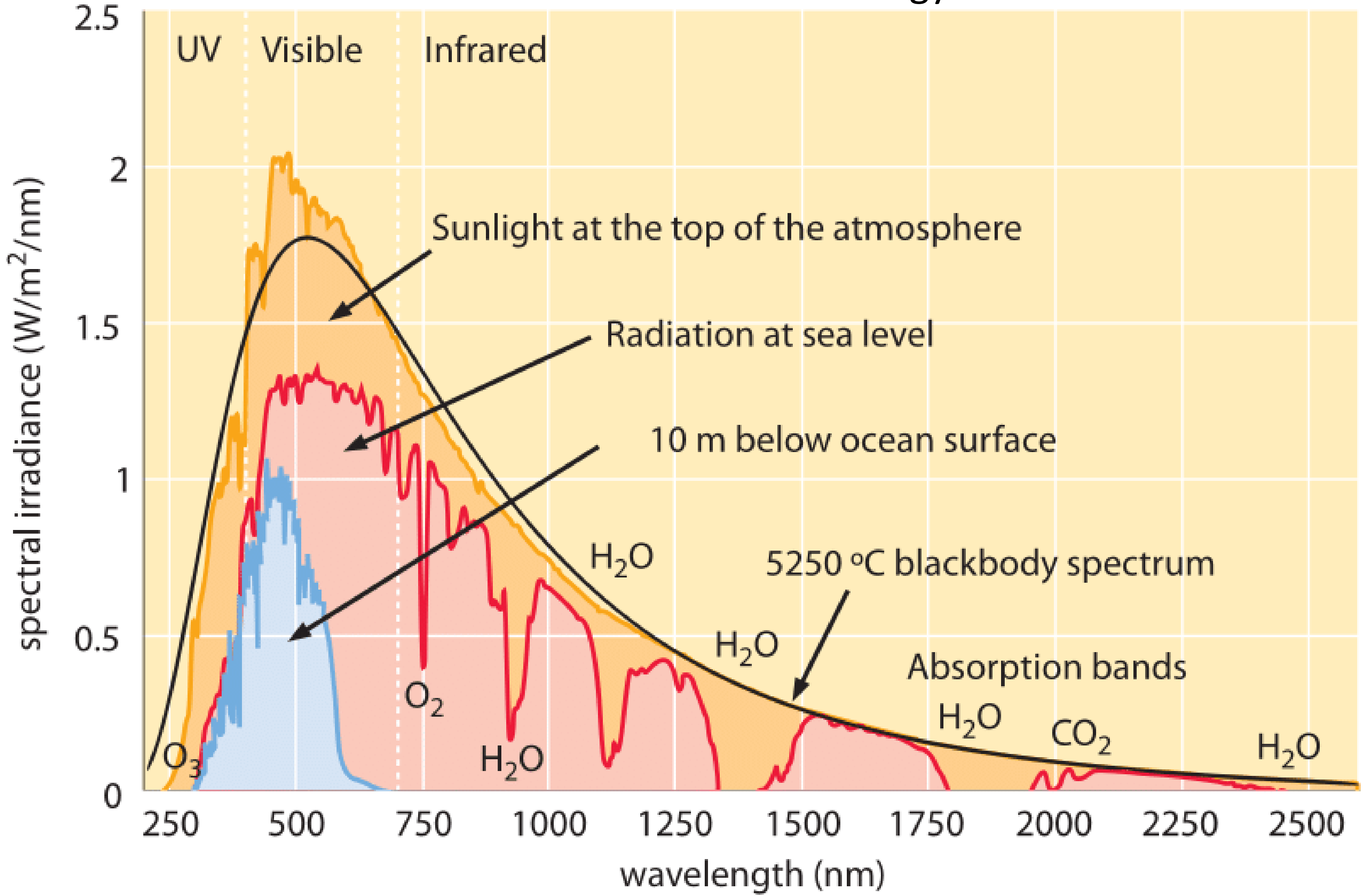
**The majority of outgoing IR comes from gases**



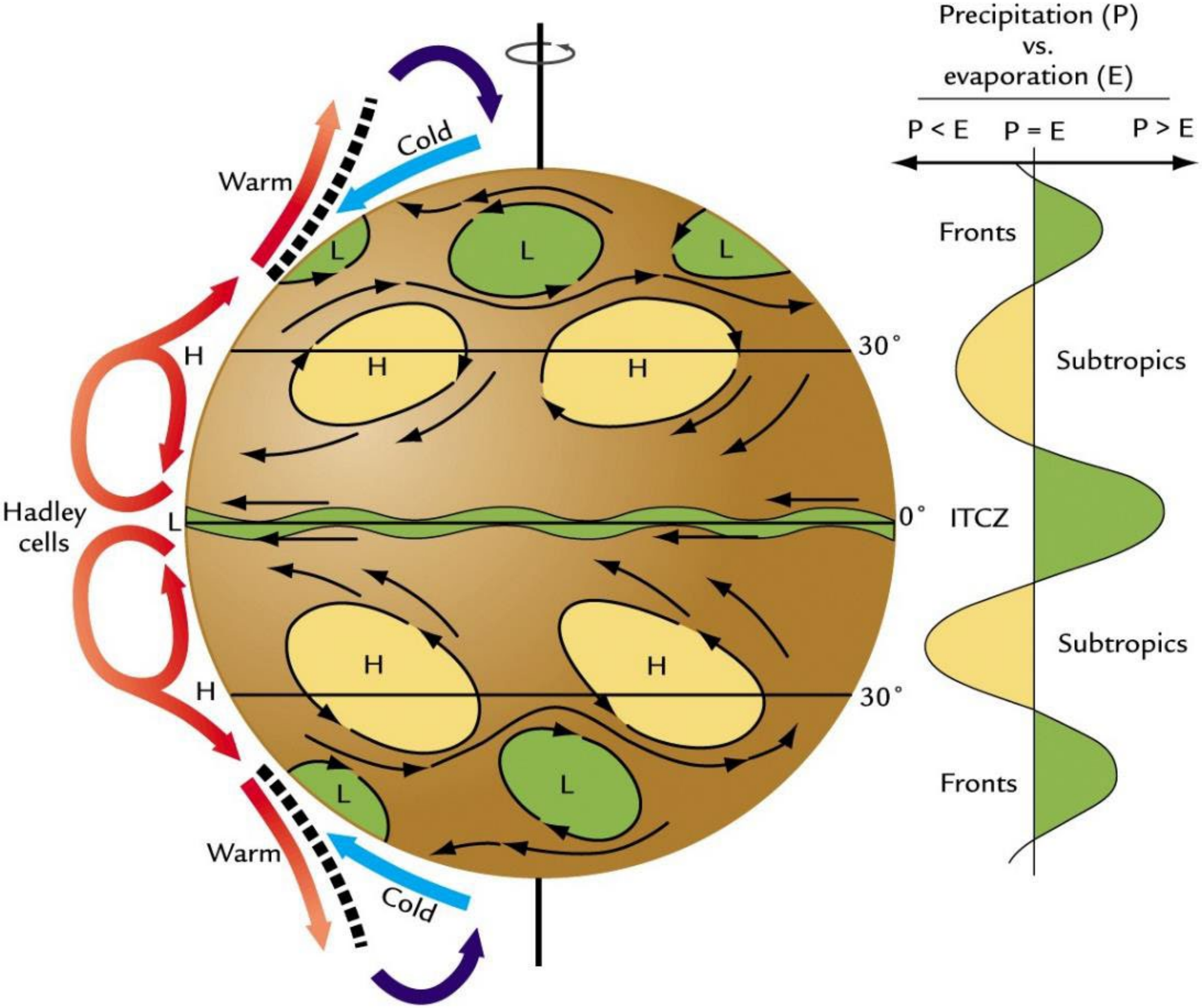

~50%

# Incoming Energy from the Sun

It takes about 8 minutes for the sun's energy to reach the Earth



# Three-cell model of atmospheric circulation

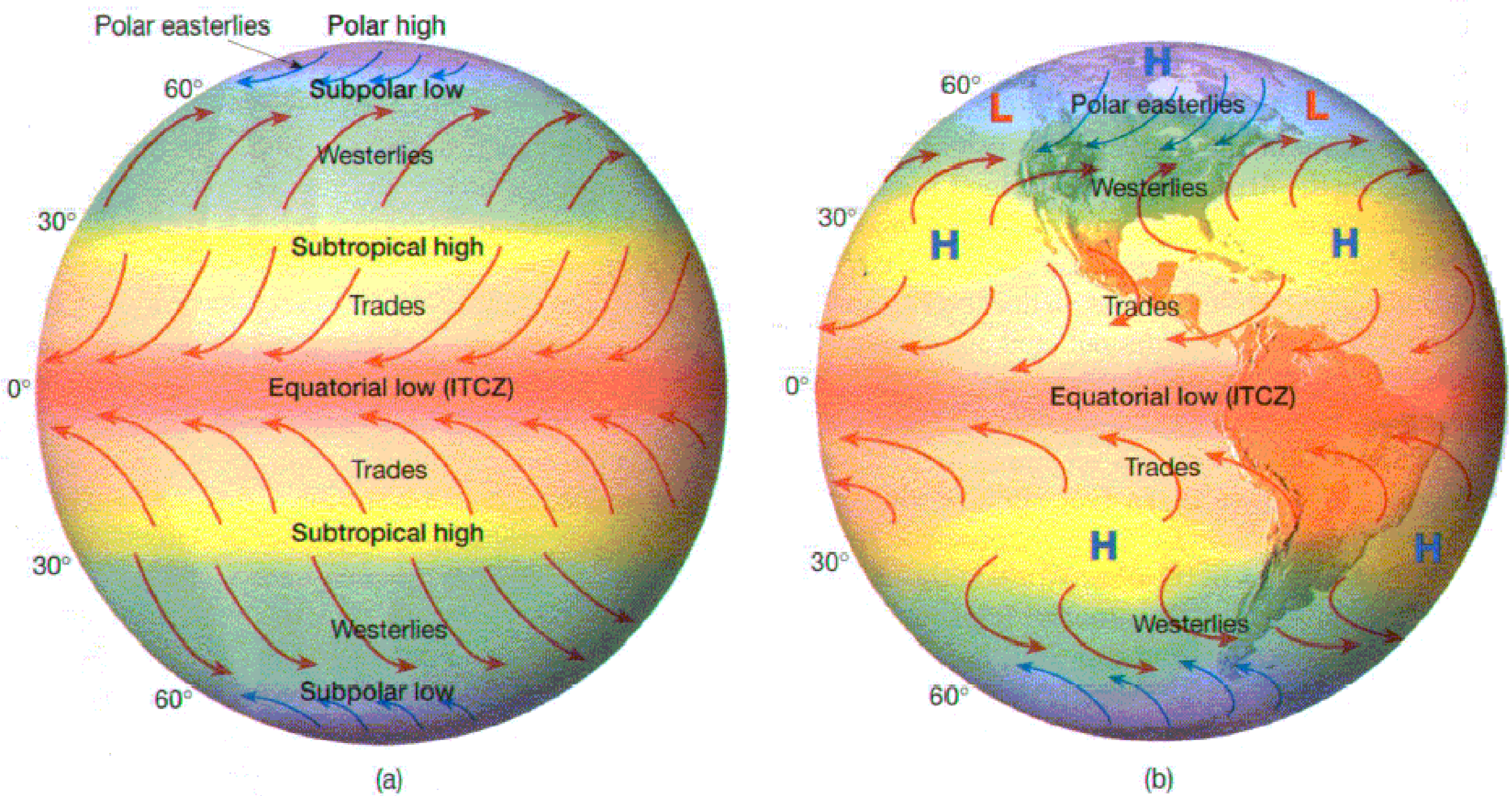



**(1) Polar cells**

**(2) Midlatitude cells**

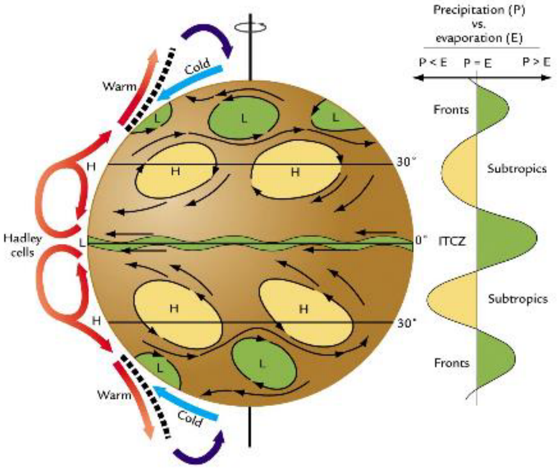
**(3) Hadley cells**

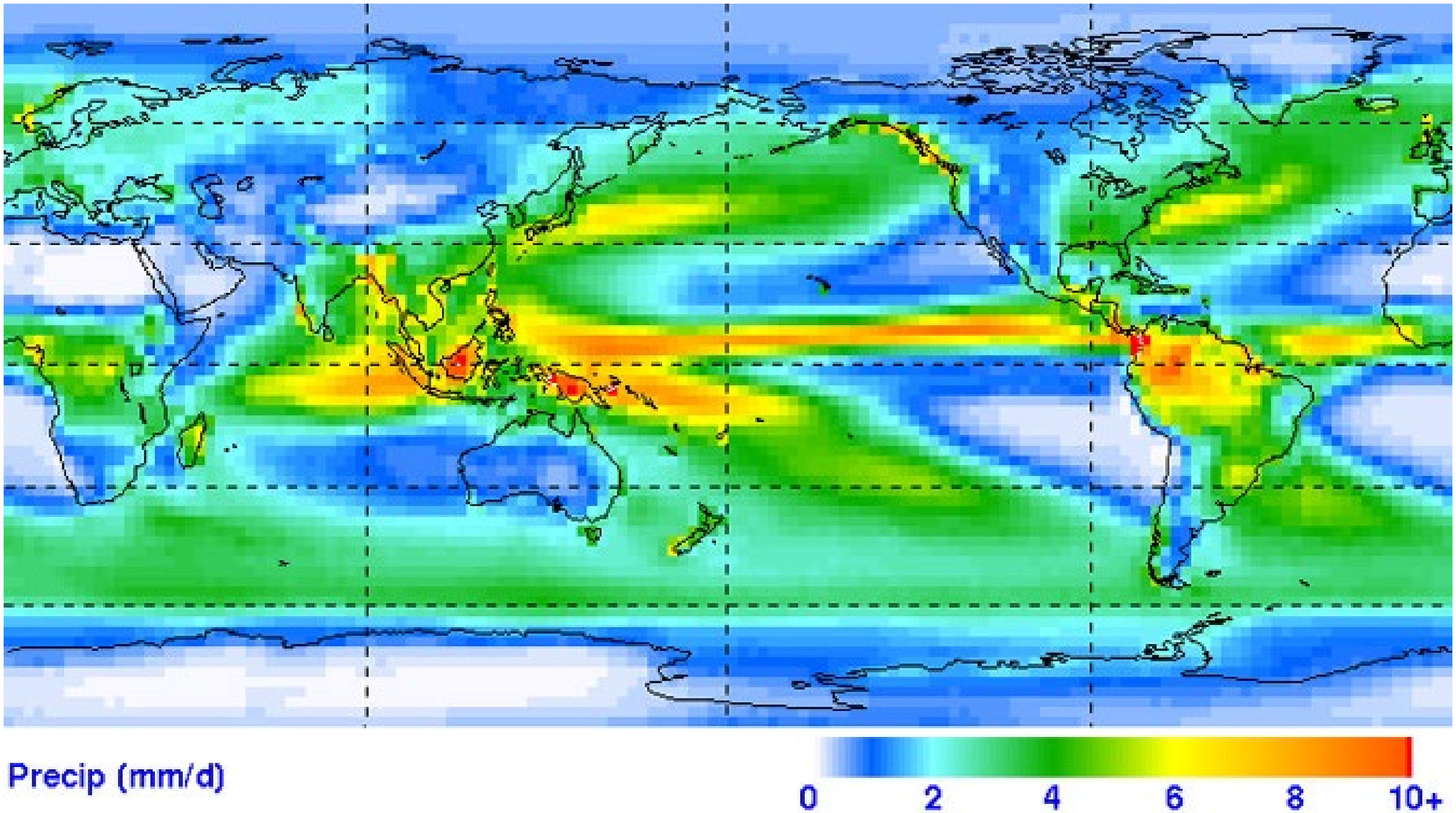
# Idealized vs. actual zonal pressure belts




**Non-uniform surface = uneven heating**
**Unstable windflow = eddies**
**Sun doesn't remain over the equator year-round = 23.5N-23.5S**

# Movement of $H_2O$ through the atmosphere determines the distribution of rainfall; global average precip ~943mm

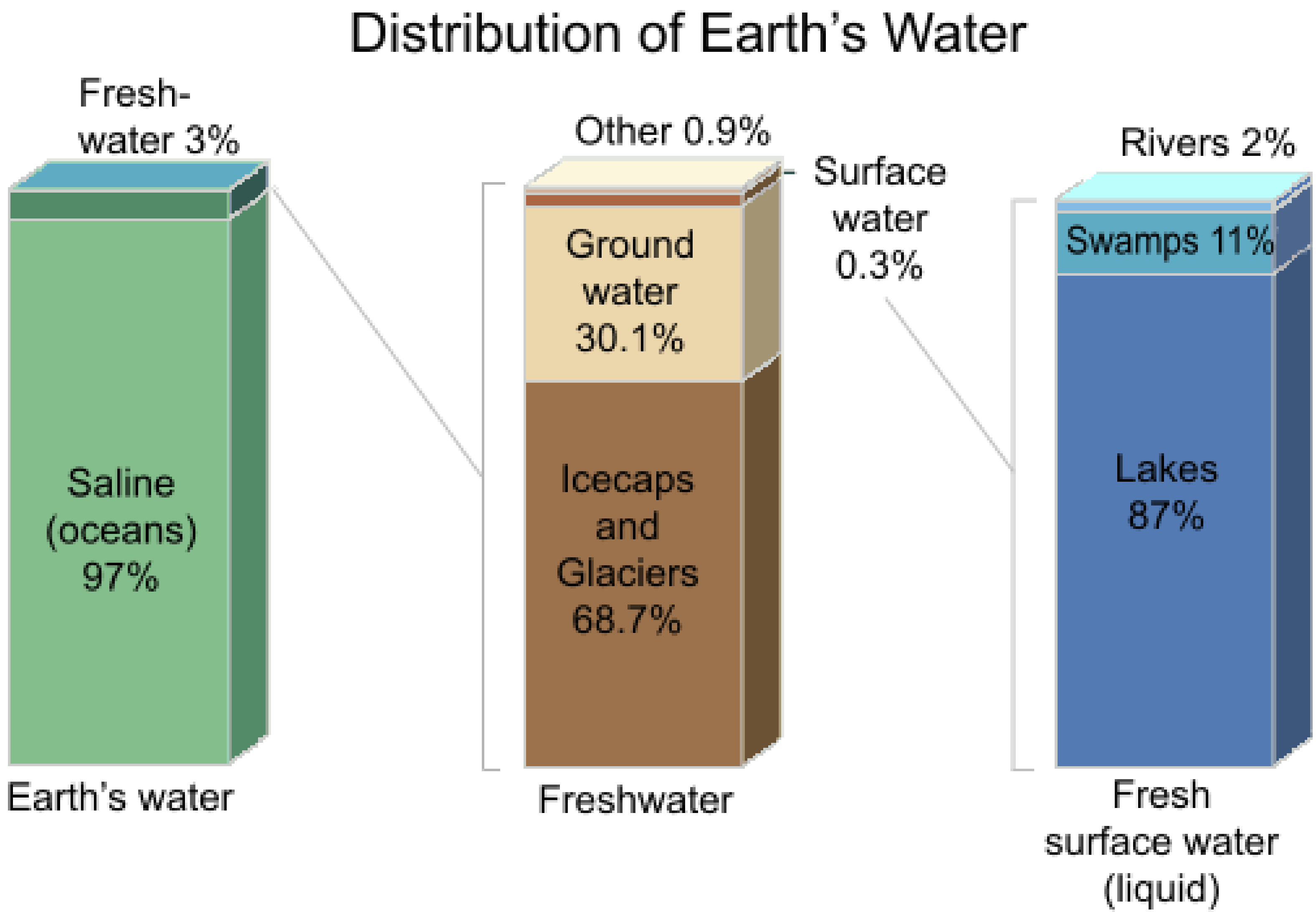


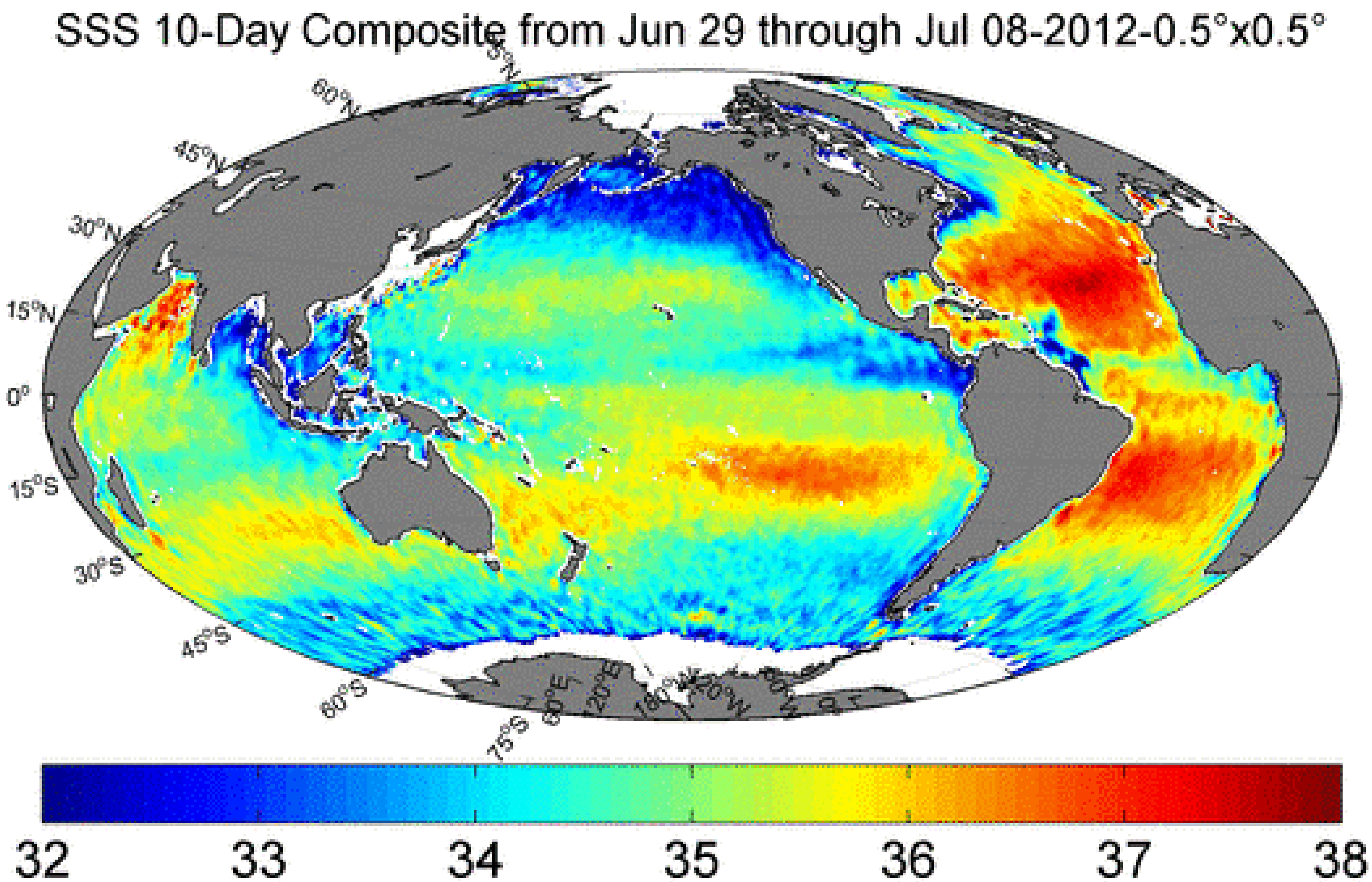

global precipitation in millimeters per day, computed from the period 1979-2008

Source: NASA

# The annual circulation of $H_2O$ is the largest movement of a chemical substance at the surface of the Earth

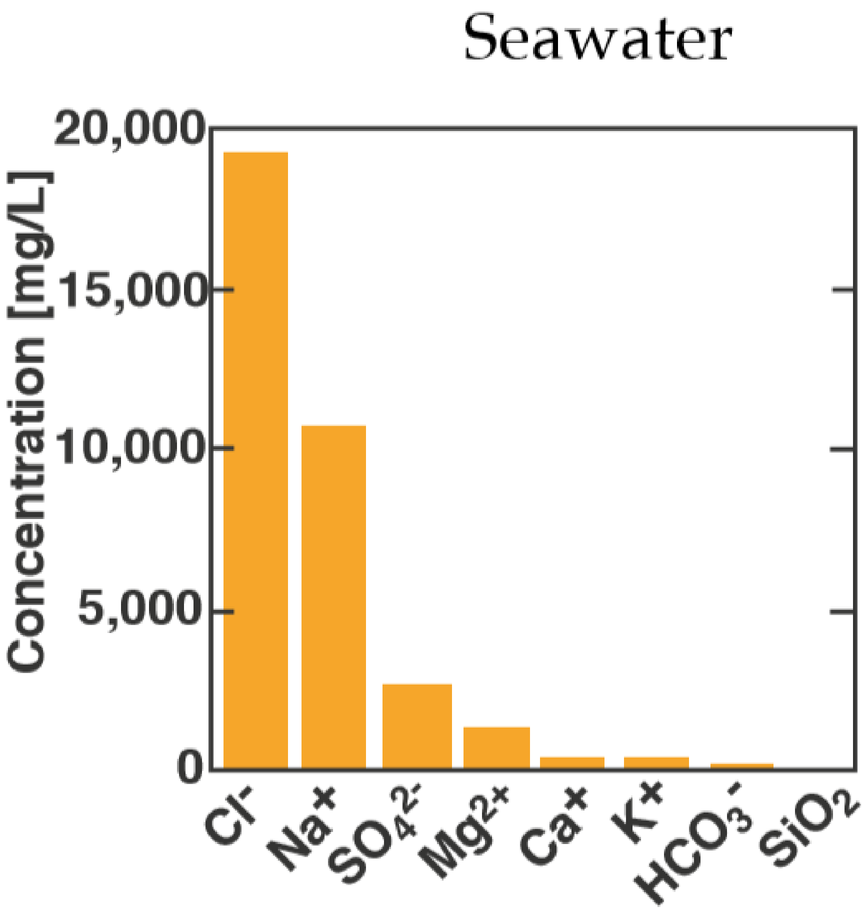


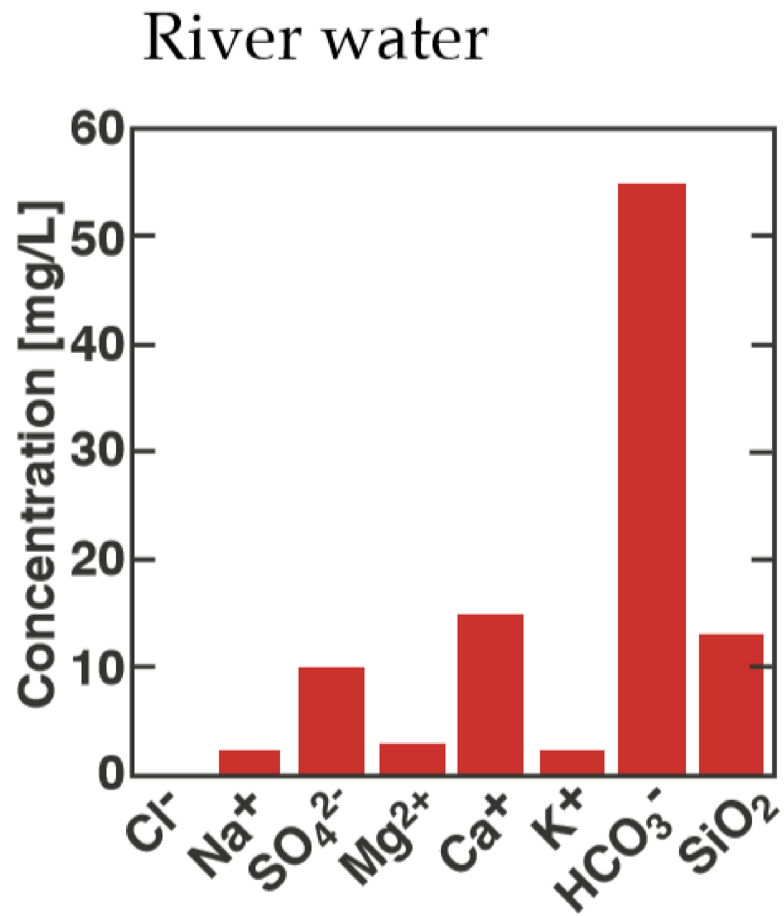
Net evaporation from the surface ocean affects surface water salinity in the ocean and increases surface water density controls thermohaline circulation of ocean



# Why is the ocean salty?

compare salt composition in the ocean to rivers



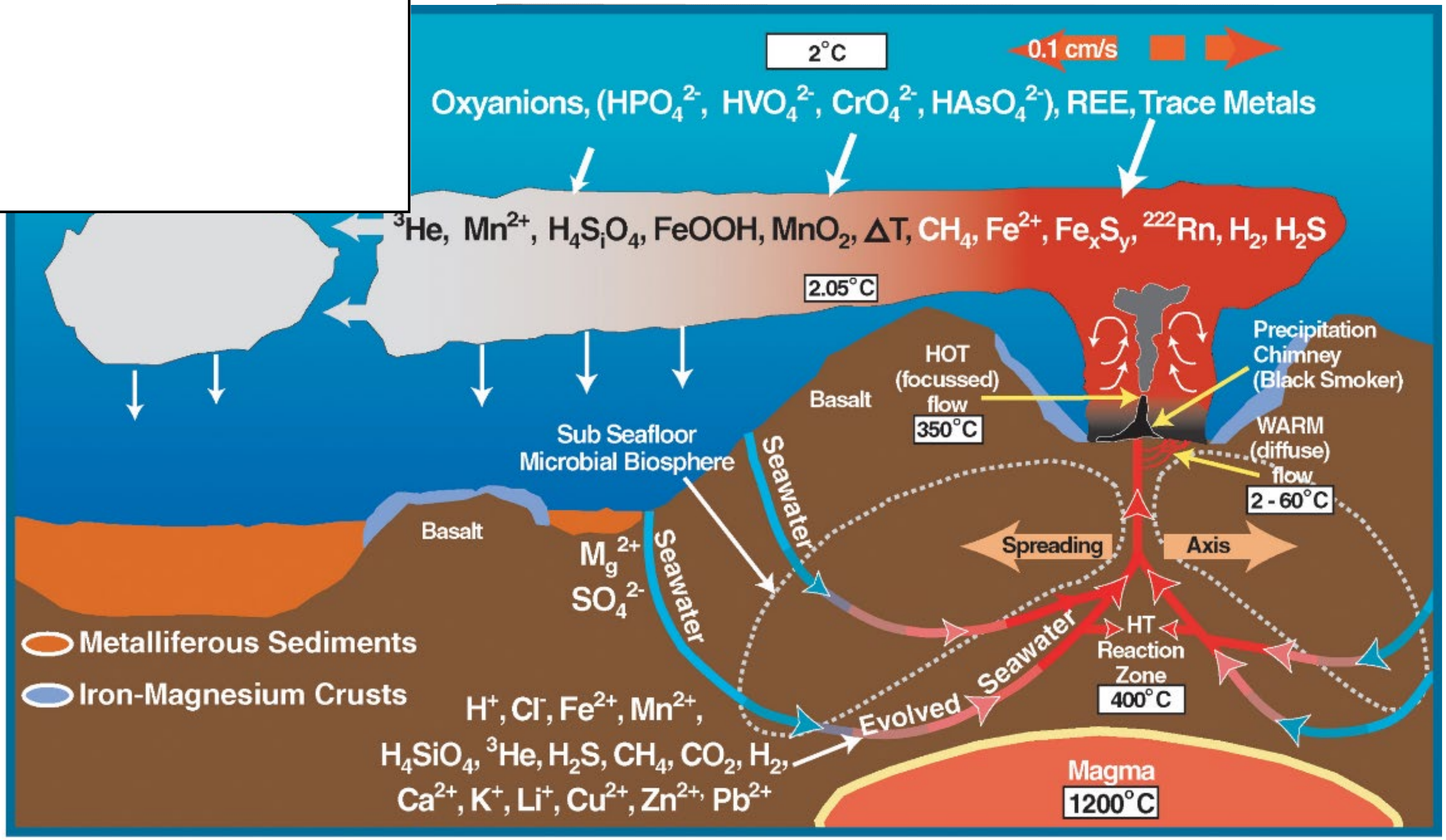

Conclusion: rivers must not be the whole story

# Excess Volatiles

Excess Volatiles: components of seawater not accounted for by rivers

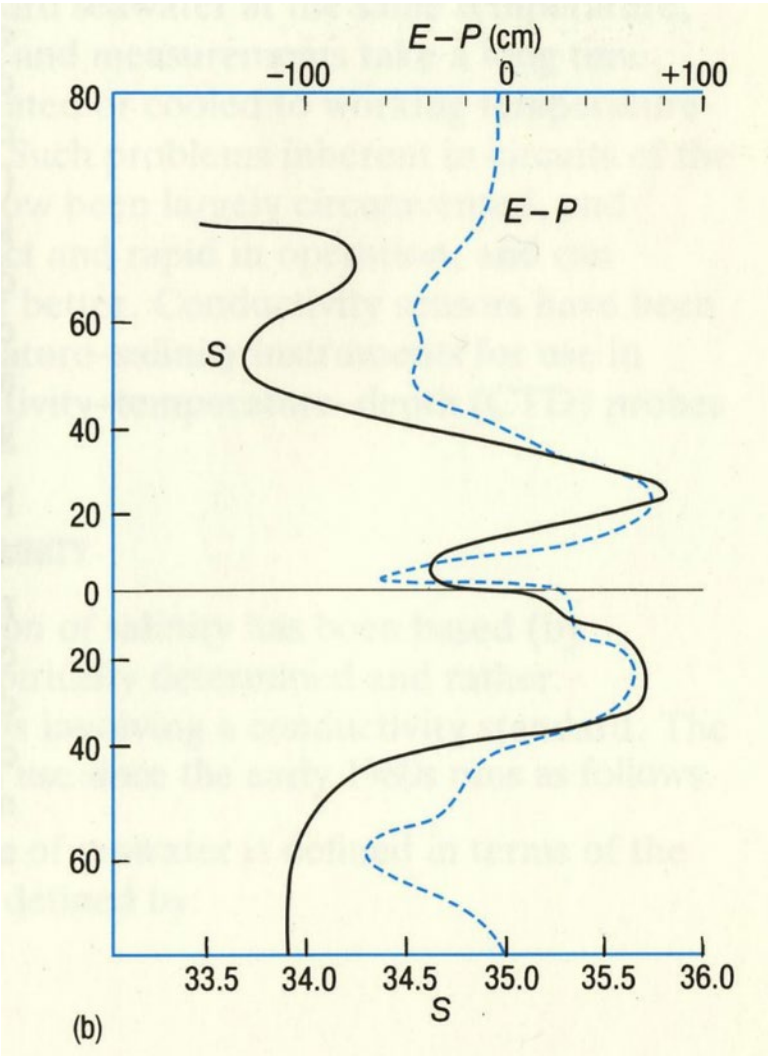
| | | |
|---|---|---|
| Chloride | $Cl^-$ | 100 million years |
| Sodium | $Na^+$ | 68 million years |
| Magnesium | $Mg^{2+}$ | 10 million years |
| Sulphate | $SO_4^{2-}$ | 10 million years |
| Potassium | $K^+$ | 7 million years |
| Calcium | $Ca^+$ | 1 million years |

Ocean chloride ($Cl^-$) primarily comes from hydrothermal activity



# Distribution of sea surface salinity

**Zonally Averaged**




from Seawater: Its Composition ..

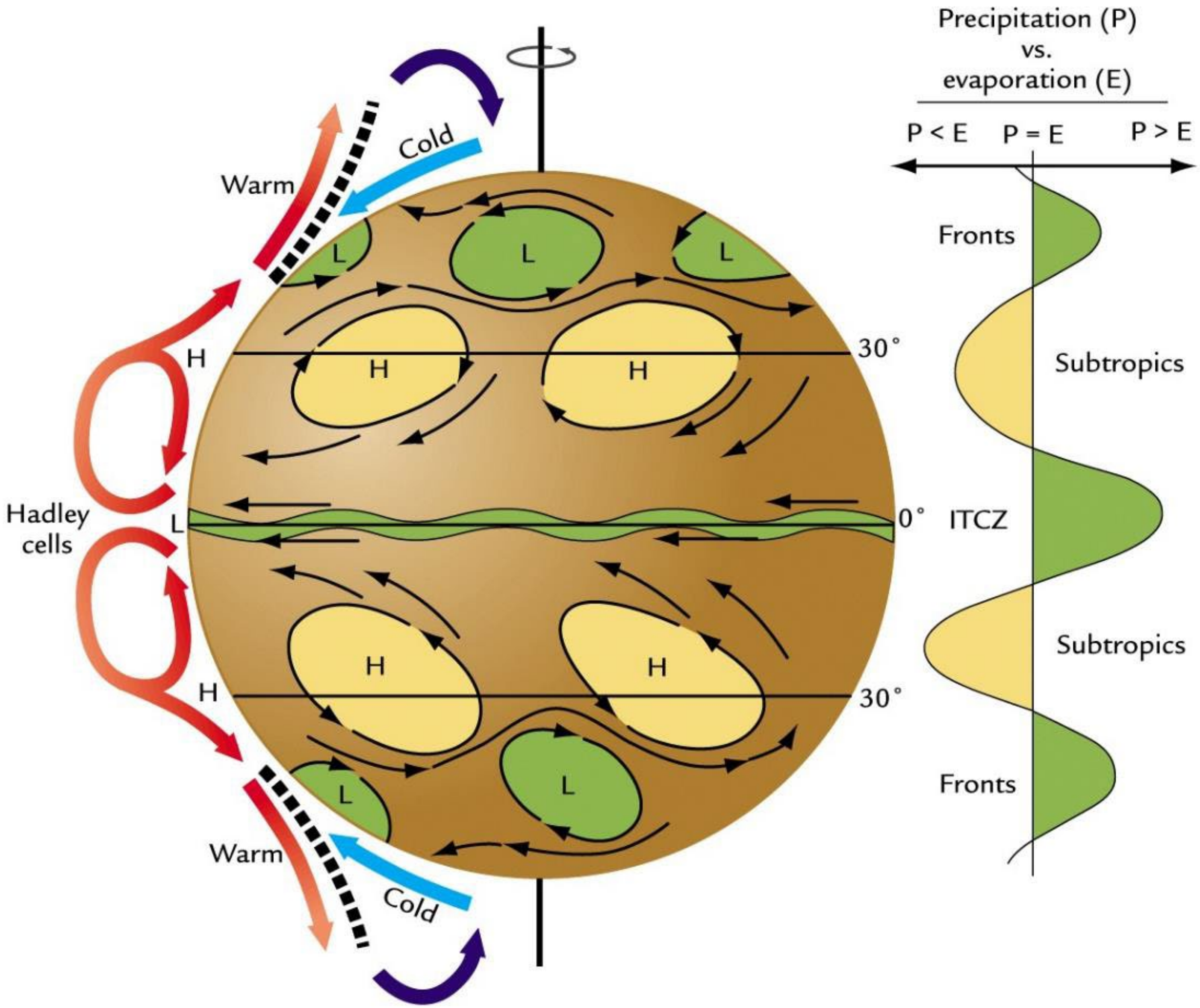
The high salinity values in the subtropics results from net evaporation.

The low salinity values in temperate regions comes from net precipitation.

The low salinity values in northern hemisphere are lower than the southern hemisphere because of river input.

High salinity values in the polar north come from seasonal sea ice formation.

# Three-cell model of atmospheric circulation

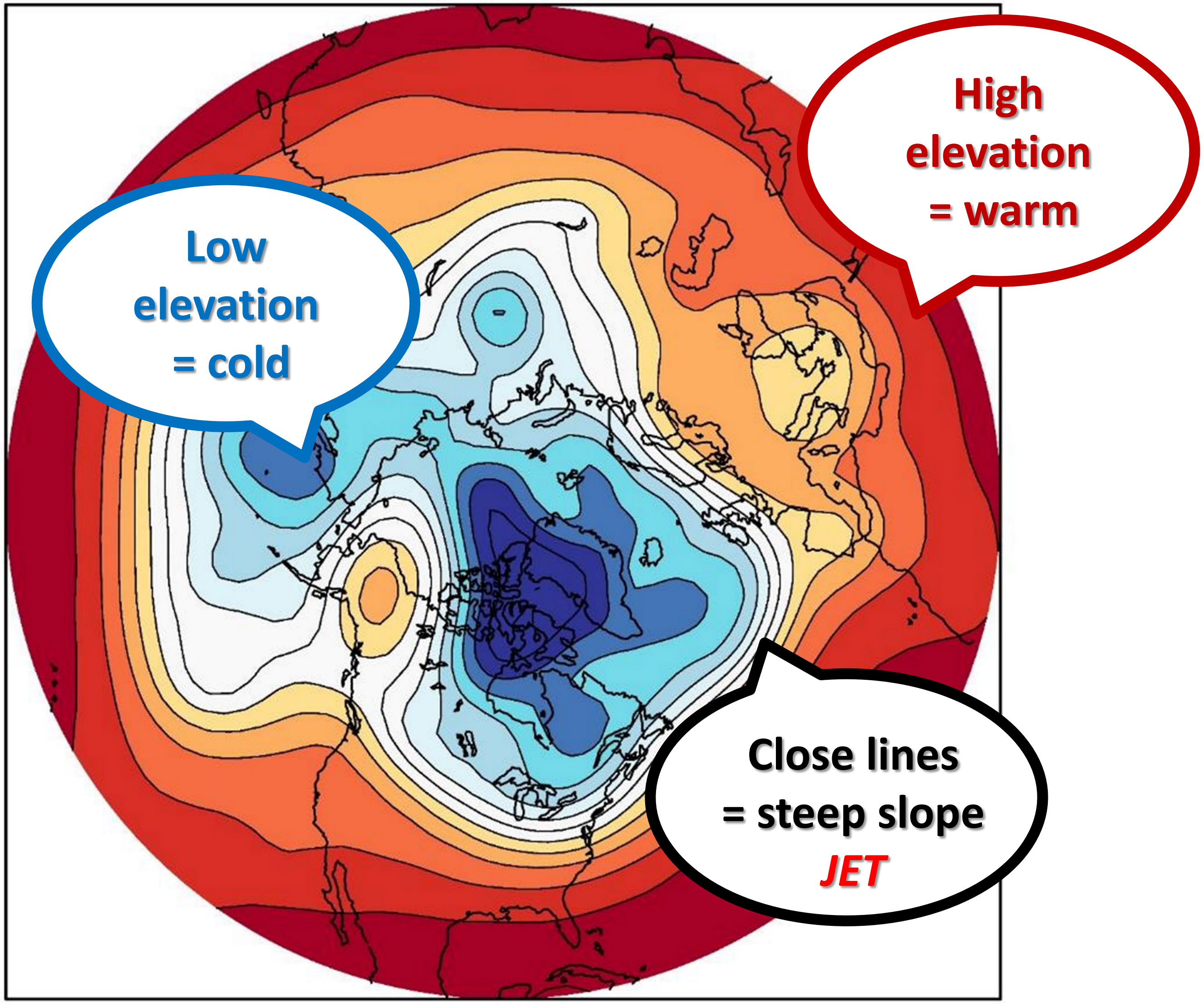



**(1) Polar cells**

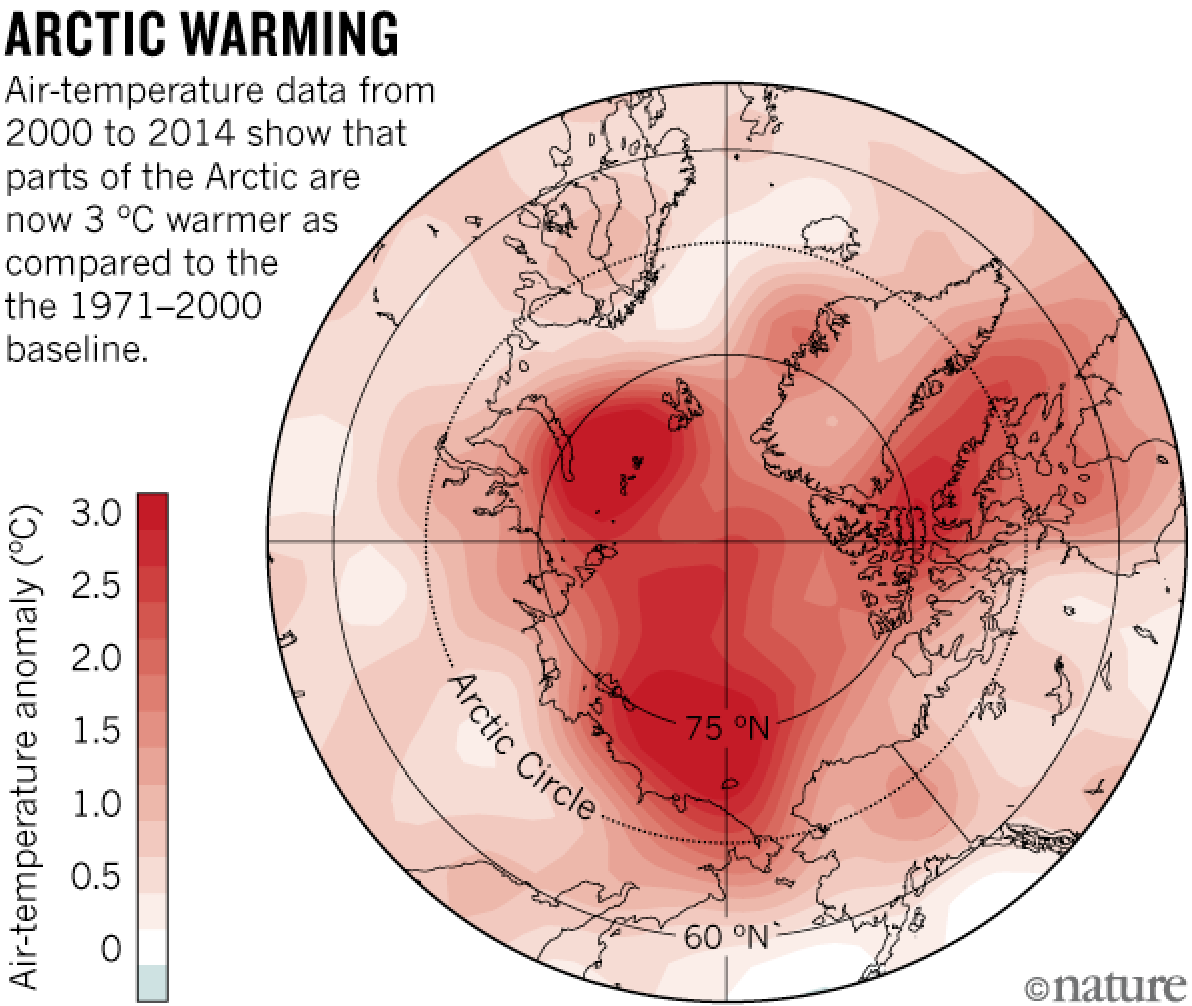
**(2) Midlatitude cells**

**(3) Hadley cells**

# A “topographic map” of a layer in the atmosphere (Height of a Constant Pressure Surface)

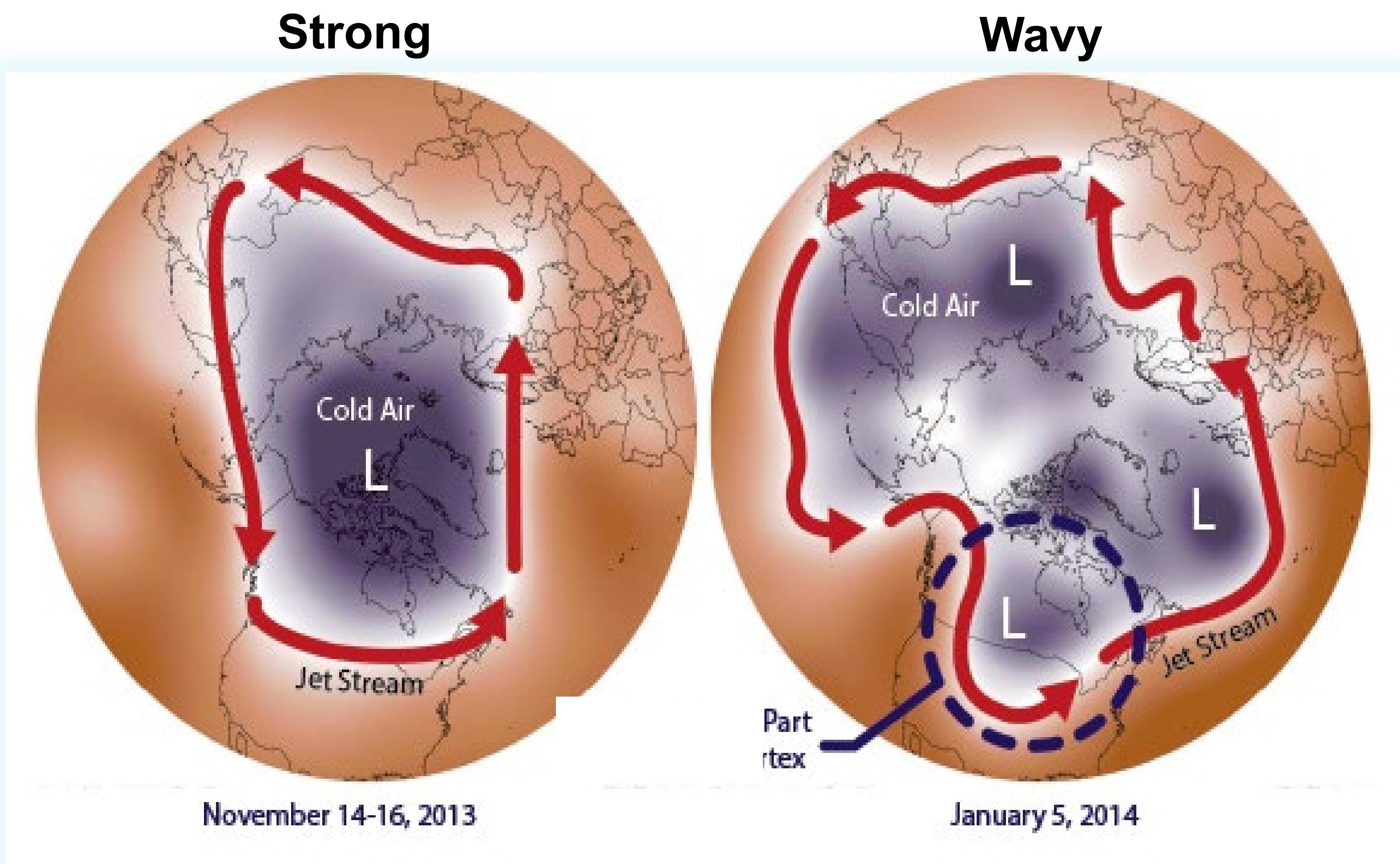


# Arctic is Warming 3x faster than Global Average



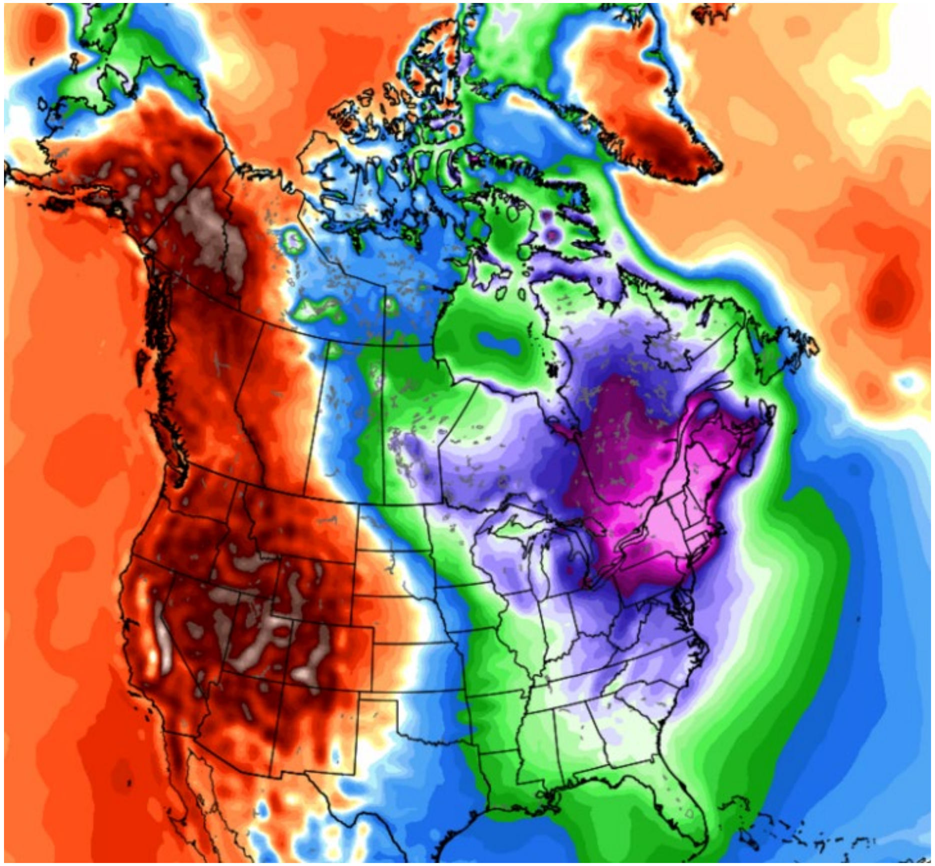

Credit: Nature, April 28, 2017, doi:10.1038/nature.2017.21911

# Different Jet Streams and (lower atmosphere) Polar Vortex

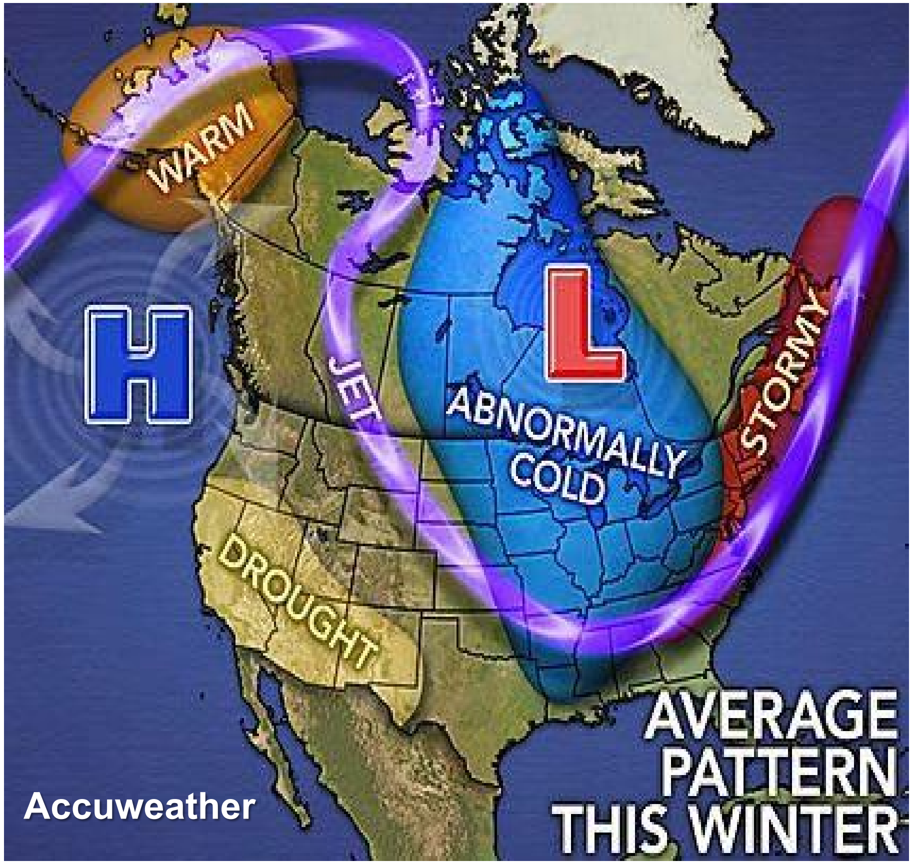


# Arctic Blasts from the Polar Vortex

February 1-14 2015



December 4-10 2016



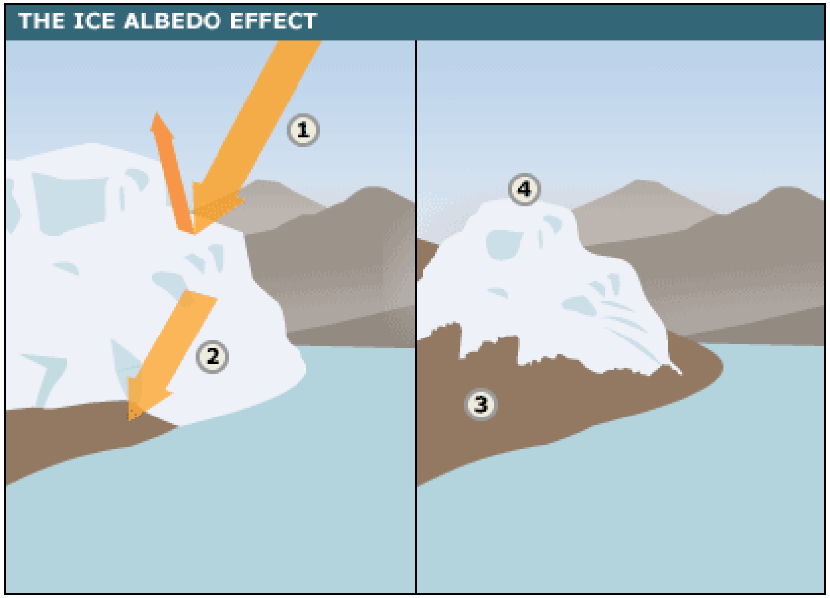
# The Cryosphere is a Critical Component of Heat Budget

1 Light colored ice reflects back the Sun's energy efficiently.

2 Exposed land is darker and absorbs more energy; warming.

3 As the ice melts, more land is exposed. This absorbs more heat, melting more ice, and causing further warming.

4 The altitude of the melting ice is reduced so it becomes harder for new ice to form (esp for Greenland).



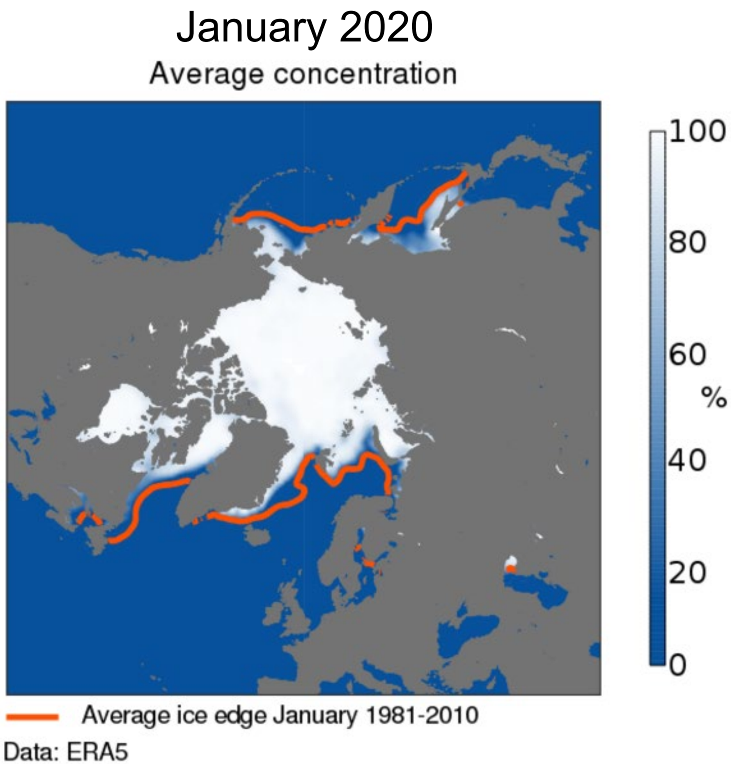

Melt water flows to the base of the Greenland ice sheet.

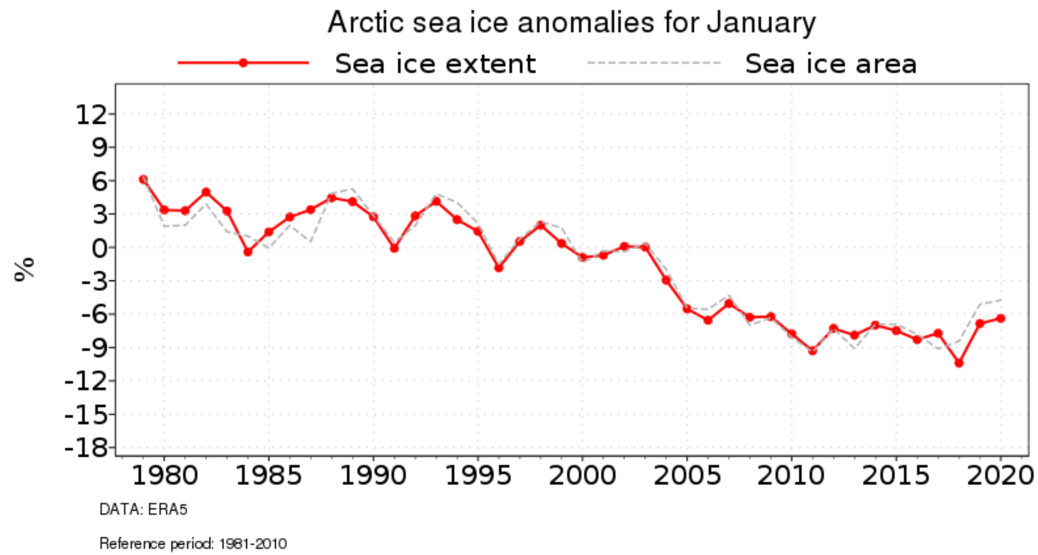
**Ice is melting MUCH faster than glaciologists had forecast**

**Positive Feedback can cause Runaway Warming, by darkening originally-light surfaces.**

The climate system is full of feedbacks

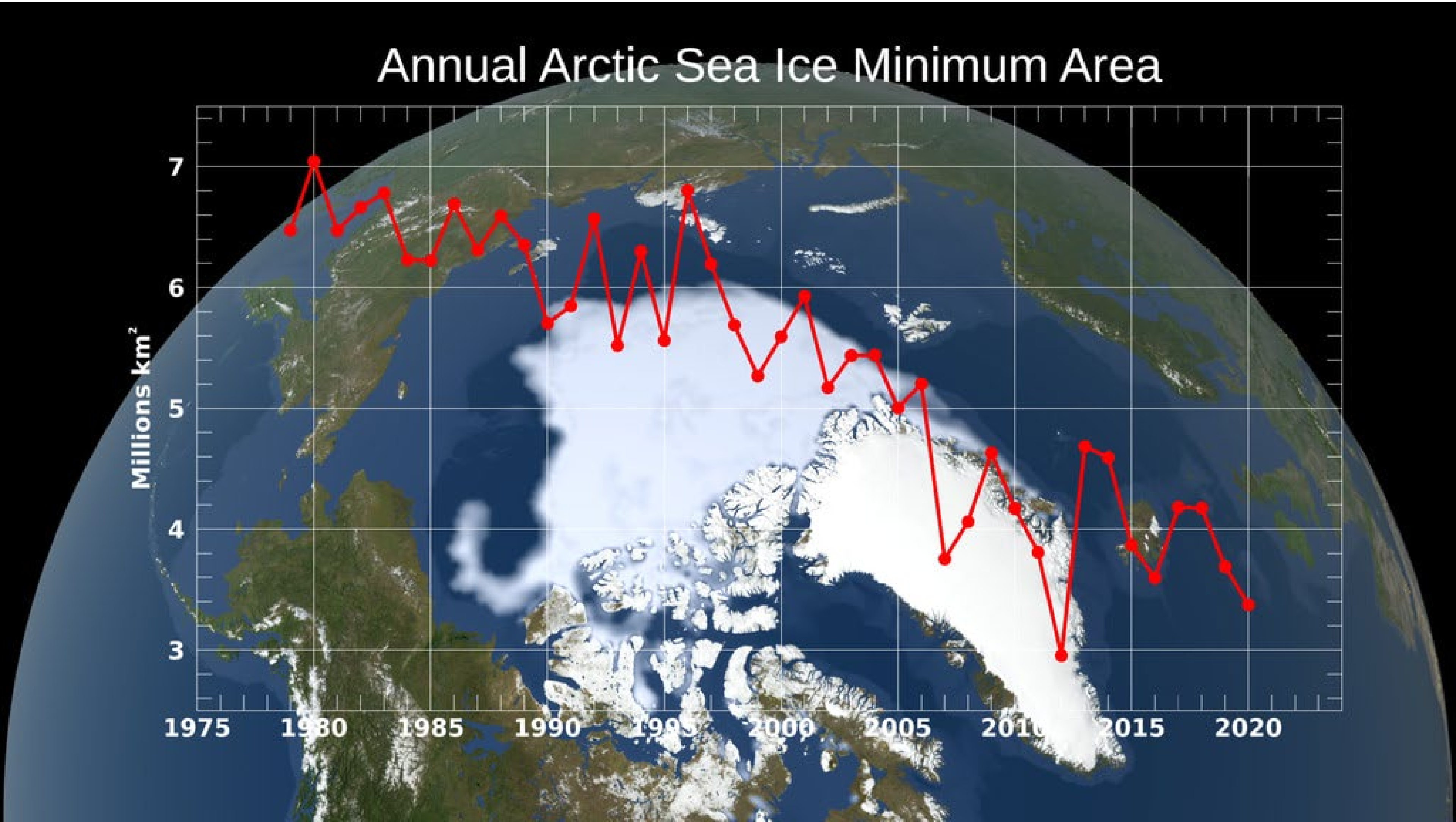
# Sea Ice Change: Small but significant changes in the winter



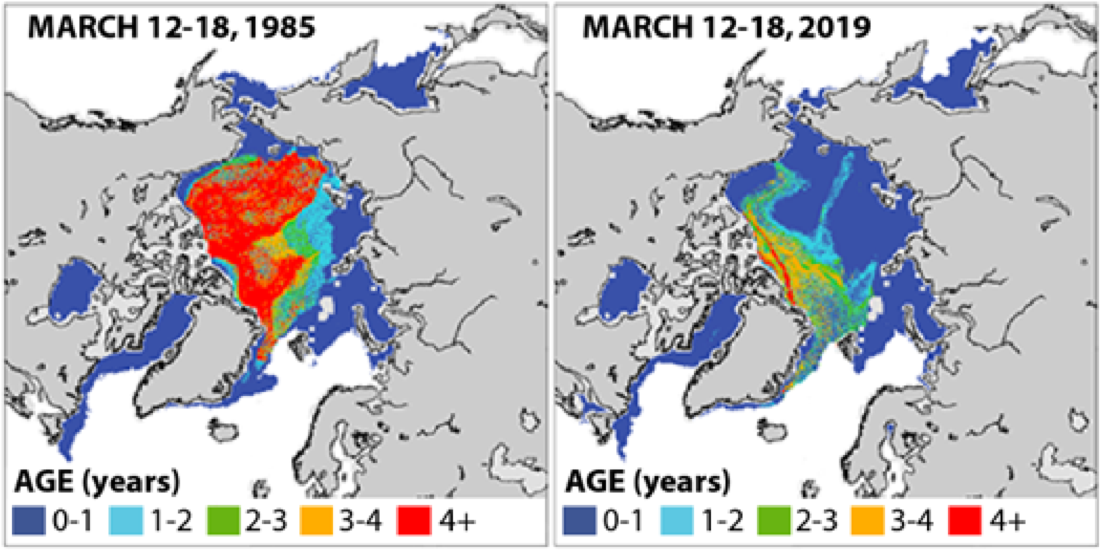

Time series of monthly mean Arctic sea ice extent (solid red) and sea ice area (dashed grey) anomalies for all January months from 1979 to 2020. The anomalies are expressed as a percentage of the January average for the period 1981-2010. Data source: ERA5. Credit: Copernicus Climate Change Service/ECMWF.

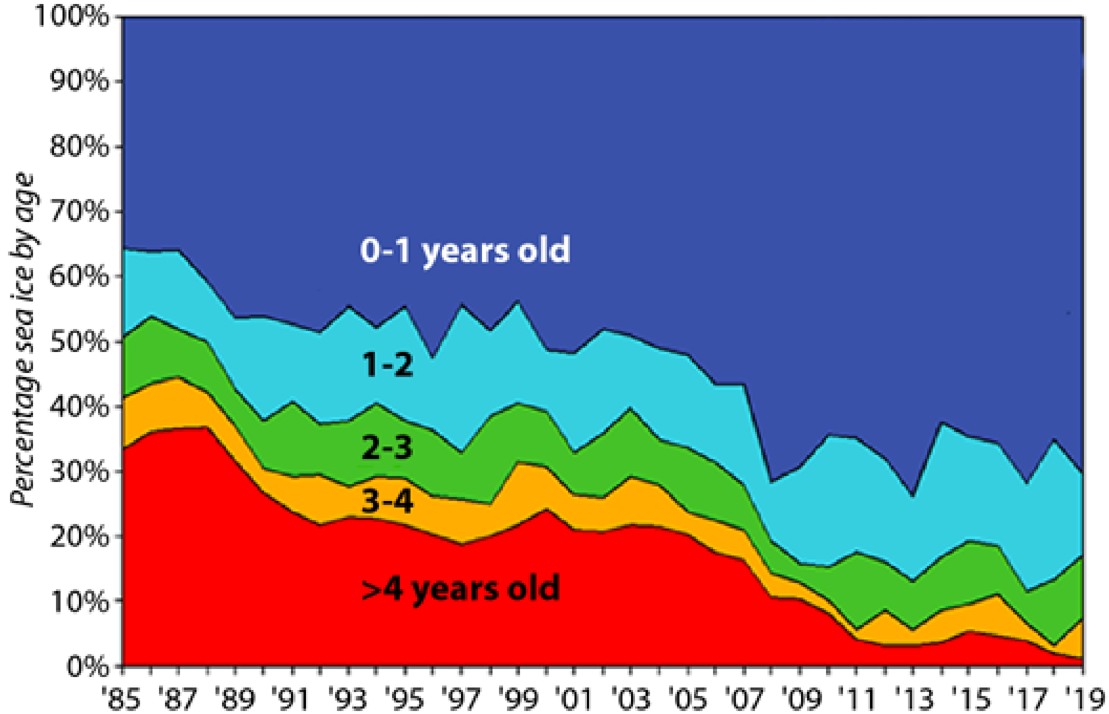
# Sea Ice Change: Much more significant changes in the summer




WORLDVIEW
Layers (7)
Hardly any sea ice left around the North Pole
Created by Sam Carana with NASA image for Arctic-news.blogspot.com
2016 SEP 08
100 km
50 mi
88.0528°, 95.5149° EPSG:3413

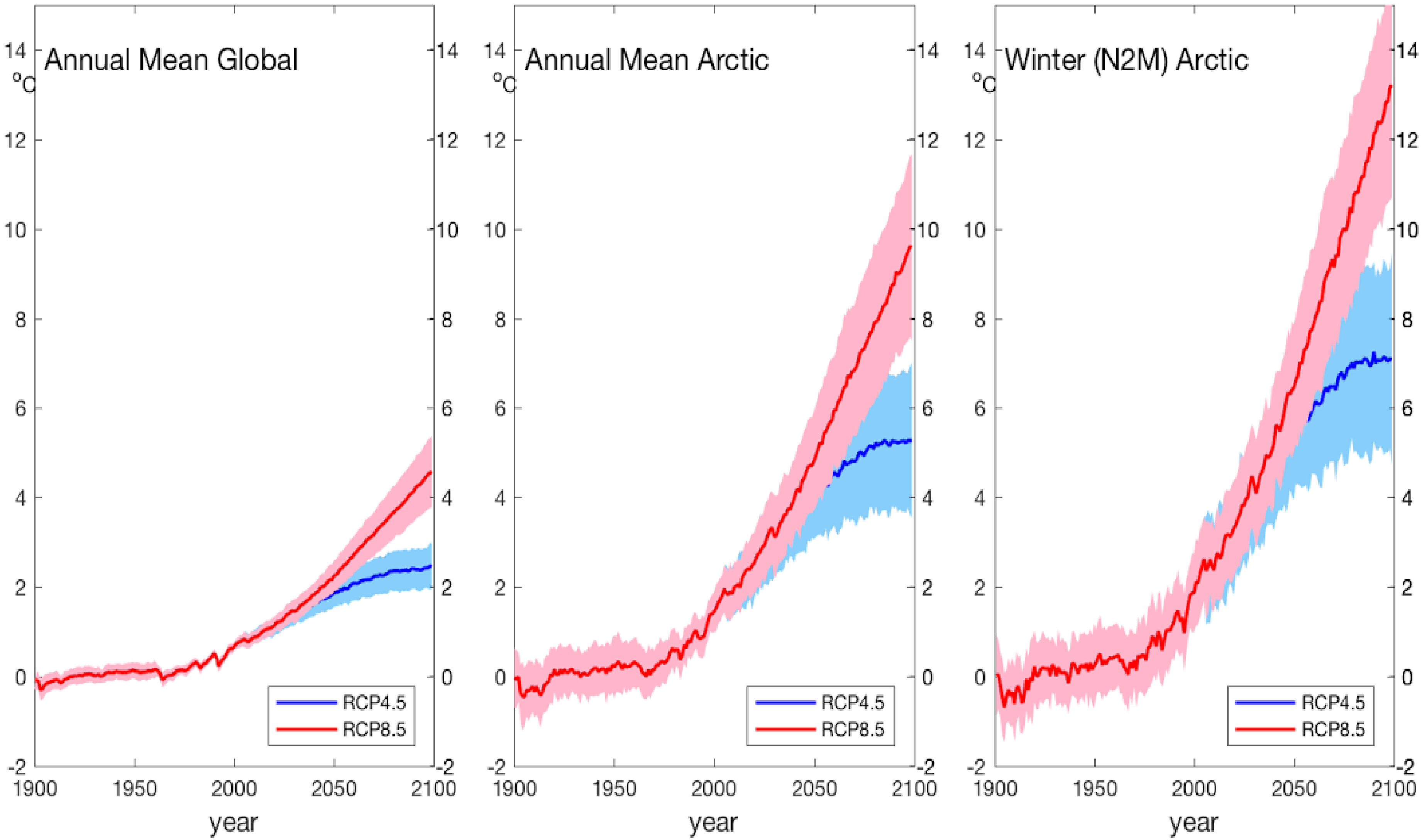
# Multi-Year Sea Ice is Disappearing



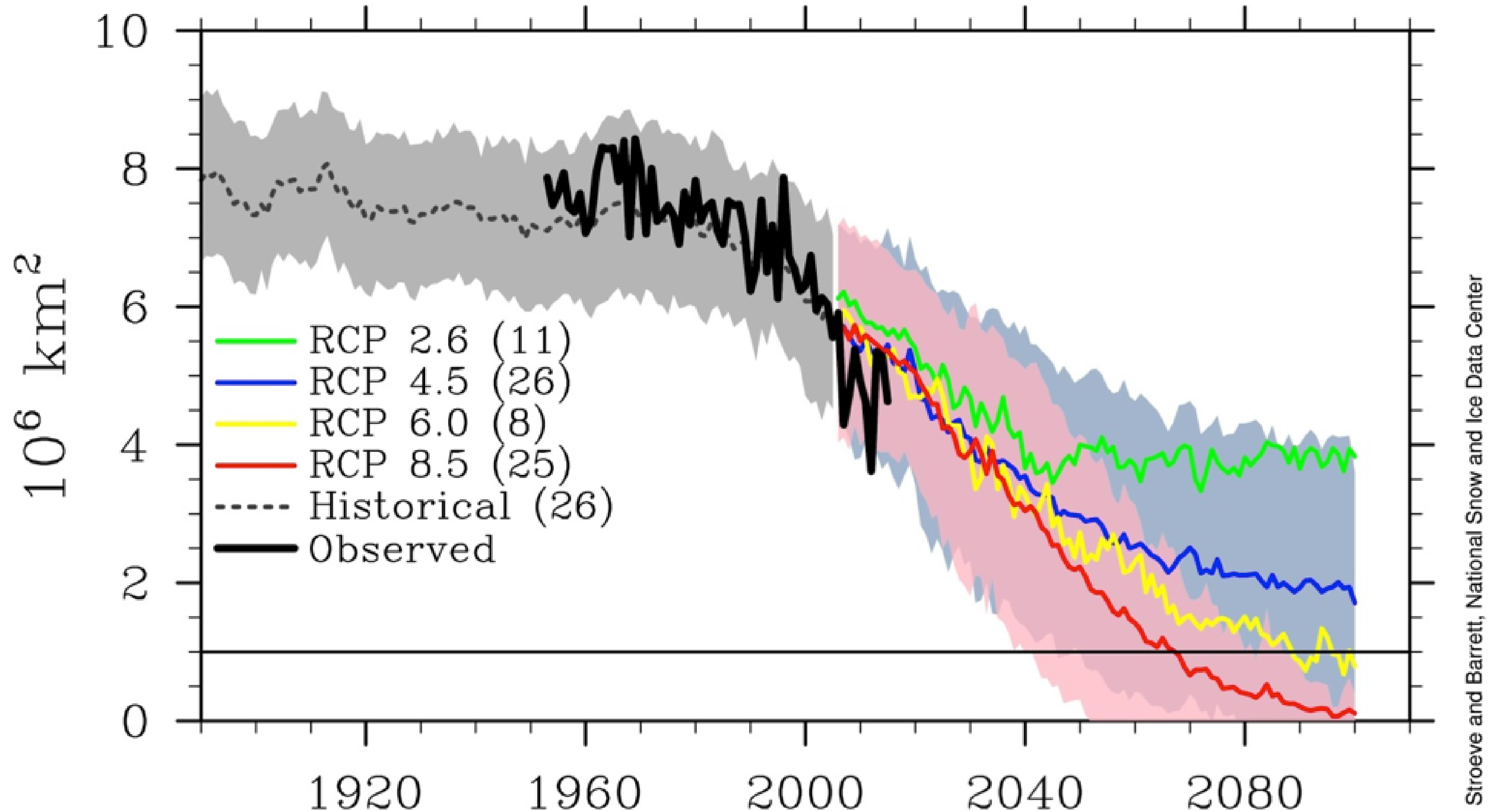




SOURCE: Arctic Report Card 2019 InsideClimate News

# Even if globe stabilizes at +2 C, Arctic does not stabilize



# Sea Ice Model Intercomparison




**Colored lines for RCP scenarios are model averages (CMIP5) and lighter shades of the line colors denote ranges among models for each scenario. Dotted gray line and gray shading denotes average and range of the historical simulations through 2005. The thick black line shows observed data for 1953-2012.**